

PAAD FUNCTIONALLITY SPECIFICATION

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Functionality specification

< The State of Arstotzka >

Version history

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Introduction

This document **provides a comprehensive specification of the functional and non-functional requirements** for the Disaster Response and Management System (DRMS). It is designed to guide the development, implementation, and verification phases for the State of Arstotzka's emergency infrastructure. The primary audience includes system architects, software developers, government procurement officers, and emergency agency stakeholders.

Scope of the project

What will be implemented: The project encompasses the design and deployment of an integrated ecosystem: a Web Command Center for strategic administration, a Mobile Tactical App for field-based responders, and an IoT Edge Network consisting of specialized sensors (seismic, hydraulic, and atmospheric). It includes the backend logic for automated alerting, encrypted data transit, and high-integrity data synchronization.

What is not part of the project: The project does not include the physical hardware manufacturing of sensors; no public alerts (the text mentions alerts for "active client devices" (responders/admins), but it does not mention a public-facing app or SMS system to warn the general population of Arstotzka).

Concepts

The Disaster Response and Management System (DRMS) is based on the concept of integrating real-time environmental monitoring, centralized decision-making, and coordinated field response. The system combines data from IoT sensors with reports from emergency responders to provide situational awareness and support effective disaster management.

Role description

Disaster Management Administrator: Acts as the strategic controller. They interpret heatmap data, authorize the mobilization of state resources (rescue teams, medical supplies), and generate post-incident audit reports to evaluate response efficiency.

Emergency Responder: Operates in affected areas and serves as the primary source of field intelligence. Responders receive alerts and task assignments via the mobile application, navigate using real-time GPS routing, and submit structured incident reports enriched with multimedia evidence.

IoT: The IoT is a non-human actor, which automatize sensors and gateways that collect and transmit environmental data without direct user intervention.

Assumptions and dependencies

The system assumes that IoT sensors are correctly installed and operational, that emergency responders and administrators have compatible devices, and that priority communication channels are available.

The operation depends on telecommunication networks for data transmission, on the continuous availability of sensor data, and on mapping and geolocation services. Offline functionality ensures the system can continue operating when these resources are temporarily unavailable.

List of requirements

Requirements related to functionality

Real-Time Environmental Monitoring and Threat Visualization (Users – Command Center)

The system **shall provide users (disaster management administrators) with real-time visualization of environmental conditions and detected threats**. The web-based command center shall display data received from IoT sensors and field reports through dashboards, charts, and heatmaps, enabling users to quickly assess disaster severity, affected areas, and evolving risks in order to support timely decision-making.

IoT Data Collection and Sensor Communication (IoT System)

The IoT system **shall continuously monitor seismic activity, flood levels, and air quality** through distributed sensors installed in disaster-prone areas. Sensors **shall transmit collected data and alerts to the central system in real time** using secure and reliable communication protocols, even under unstable network conditions.

Incident Reporting by Emergency Responders (Emergency Responders)

The mobile application **needs to allow emergency responders to report incidents directly from the field**. Responders **shall be able to submit textual descriptions, photos, videos, and observations** related to disaster conditions, ensuring that the command center receives up-to-date situational information.

Geolocation and Navigation Support (Emergency Responders)

The system shall provide geolocation services for emergency responders through the mobile application. Responders shall be able to view their own location, affected areas, and receive navigation assistance to reach disaster sites efficiently.

Secure Priority Communication Channels (Emergency Responders & Users)

The system **shall enable direct communication between emergency responders and the command center** via in-app messaging. This communication shall be conducted through priority and reserved channels provided by telecom operators, ensuring reliable coordination in compliance with emergency communication regulations.

Resource Deployment and Coordination (Users – Command Center)

The command center application **shall allow users to deploy and coordinate resources** such as rescue teams, medical supplies, and equipment. The system shall display the status and location of deployed resources in real time to support effective operational control.

Reporting and Post-Disaster Analysis (Users – Command Center)

The system **shall generate analytical reports** after disaster events. These reports shall allow users to review response actions, resource utilization, and response effectiveness, supporting continuous improvement of disaster preparedness and management strategies.

Requirements related to characteristics

Usability

The system needs to support multiple languages to ensure accessibility for diverse teams. The mobile application shall be available in English, Spanish, Arabic, and Hindi, while the web-based command center shall support English and Spanish.

Security

All communication between IoT devices, the mobile application, and the web system **shall be encrypted** using standard protocols. The web application shall require secure authentications for all users, and administrators shall use multi-factor authentication. Sensitive data, including team locations, incident reports, and victim information, shall be protected against unauthorized access.

Performance

The system **shall process and display data from IoT sensors in real time**, with a maximum delay of two seconds between data capture and visualization. The mobile application shall receive and display emergency alerts instantly, even under low or intermittent connectivity conditions. The system shall support at least 5,000 simultaneous users across web and mobile applications without significant performance degradation.

Reliability

The system supports offline operation for both the mobile and web applications, as well as for IoT devices where applicable. Data collected while offline **shall be stored locally and automatically synchronized** with the central server once network connectivity is restored.